

1.. Introduction (Background and Motivation)

Operating systems are essential software that manage computer hardware and provide services for users and applications. They make it possible to run programs, manage files, and control system resources efficiently. Different operating systems are designed with different goals such as performance, stability, and security.

One of the operating systems used in this project is OpenIndiana, which is a Unix-based operating system derived from OpenSolaris. It is known for its stability, advanced filesystem support like ZFS, and strong performance in system-level operations. Although it is not as commonly used as Windows or Linux, it is important for understanding UNIX-based systems and their architecture.

The development of virtualization technology has made it easier to experiment with different operating systems without affecting the main computer. Using tools like VirtualBox or VMware, it is possible to install and run OpenIndiana in a virtual environment. This approach is safe and flexible, especially for students who want to practice installation and system configuration.

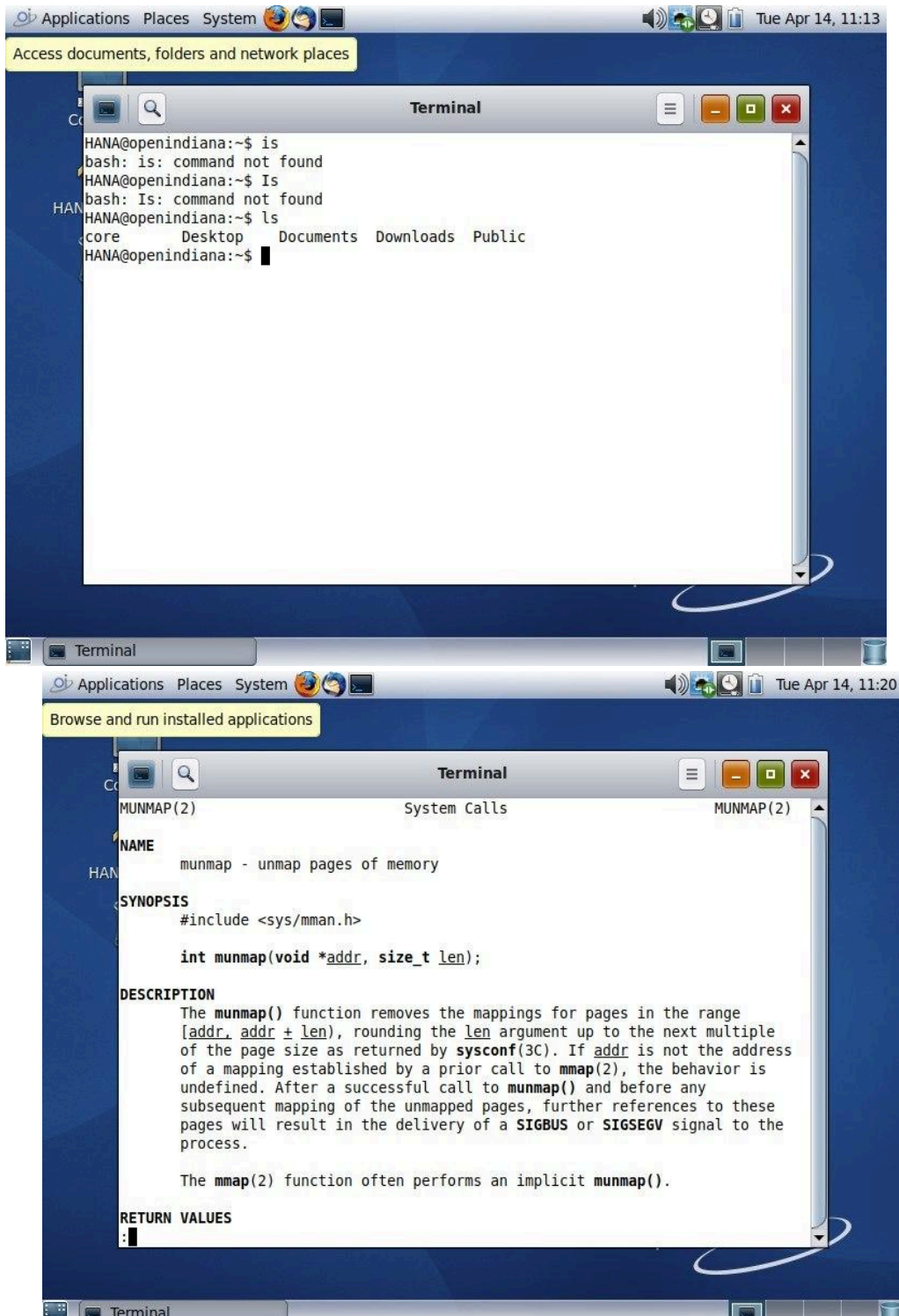
The main motivation of this project is to gain practical knowledge about installing and working with OpenIndiana in a virtual machine. It helps to understand how operating systems function in real situations, not just in theory. During the process, different challenges may occur, and solving them improves problem-solving skills and technical confidence.

- OpenIndiana tries to follow POSIX while maintaining Solaris features
- Developers must handle these differences carefully

11. System call implementation

System calls are the interface between user programs and the operating system. In OpenIndiana, system calls allow applications to request services from the kernel, such as file handling, process control, and memory management.

In OpenIndiana, system calls are implemented in a layered way. When a user program wants to perform an operation (for example, reading a file), it does not directly access the hardware. Instead, it sends a request to the operating system through a system call.



```
mouse, instance #0
pit_beep, instance #0
display, instance #0
pci8086,1e, instance #0
pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
input, instance #0
pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
disk, instance #0
cdrom, instance #1
fw, instance #0
cpu (driver not attached)
cpu (driver not attached)
sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HANA@openindiana:~$
```

```
mouse, instance #0
pit_beep, instance #0
display, instance #0
pci8086,1e, instance #0
pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
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pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
disk, instance #0
cdrom, instance #1
fw, instance #0
cpu (driver not attached)
cpu (driver not attached)
sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HANA@openindiana:~$ vmstat
```



```

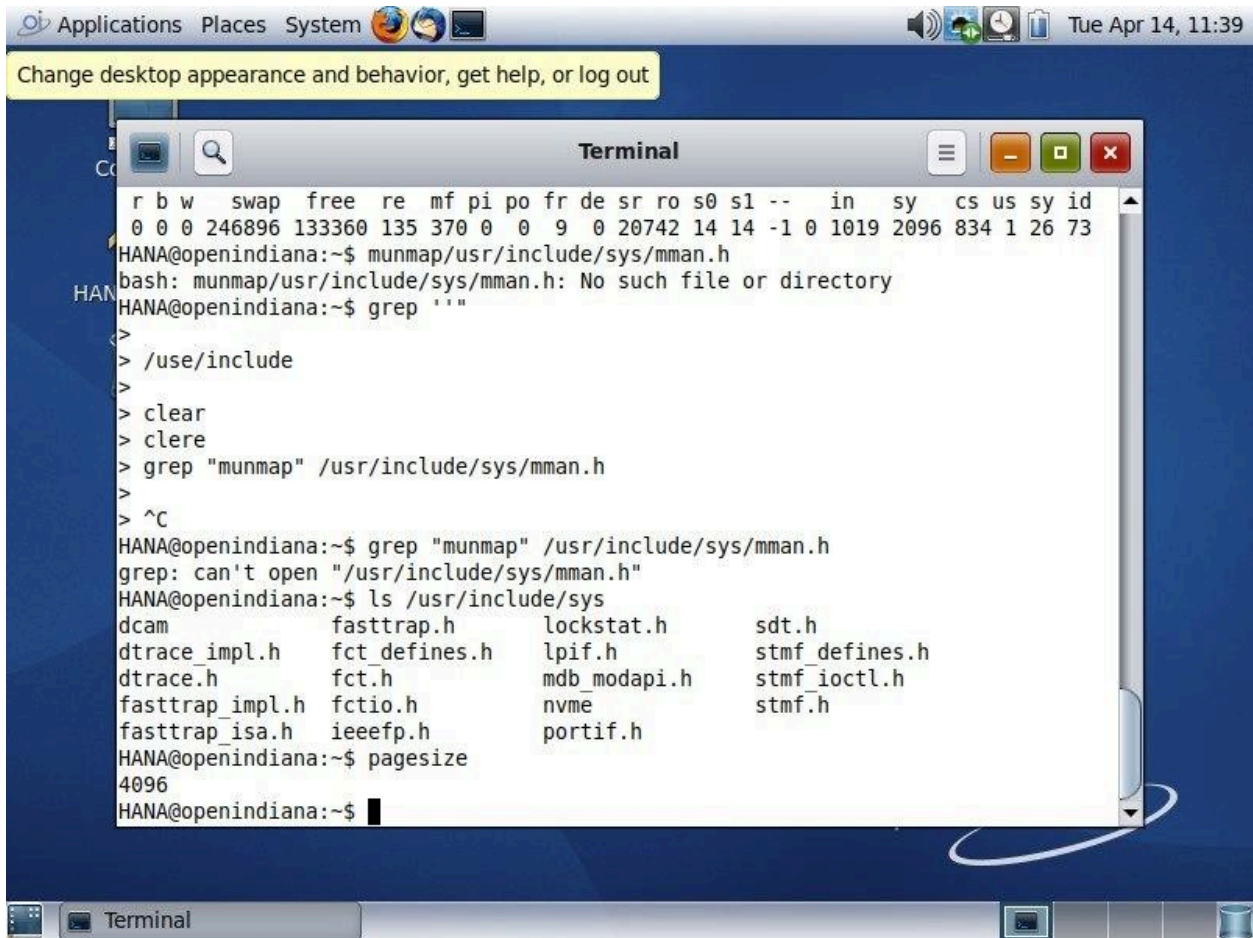
pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
    input, instance #0
pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
    disk, instance #0
    cdrom, instance #1
fw, instance #0
    cpu (driver not attached)
    cpu (driver not attached)
    sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HAN@openindiana:~$ vmstat
kthr      memory          page            disk           faults        cpu
r  b w   swap free  re  mf pi po fr de sr ro s0 s1 --  in  sy  cs us sy id
0  0  0 246896 133360 135 370 0  0  9  0 20742 14 14 -1 0 1019 2096 834 1 26 73
HAN@openindiana:~$

```

```

HAN@openindiana:~$ vmstat
kthr      memory          page            disk           faults        cpu
r  b w   swap free  re  mf pi po fr de sr ro s0 s1 --  in  sy  cs us sy id
0  0  0 246896 133360 135 370 0  0  9  0 20742 14 14 -1 0 1019 2096 834 1 26 73
HAN@openindiana:~$ munmap/usr/include/sys/mman.h
bash: munmap/usr/include/sys/mman.h: No such file or directory
HAN@openindiana:~$ grep ''
>
> /usr/include
>
> clear
> clere
> grep "munmap" /usr/include/sys/mman.h
>
> ^C
HAN@openindiana:~$ grep "munmap" /usr/include/sys/mman.h
grep: can't open "/usr/include/sys/mman.h"
HAN@openindiana:~$ ls /usr/include/sys
dcam          fasttrap.h    lockstat.h    sdt.h
dtrace_impl.h fct_defines.h lpif.h         stmf_defines.h
dtrace.h      fct.h         mdb_modapi.h  stmf_ioctl.h
fasttrap_impl.h fctio.h       nvme           stmf.h
fasttrap_isa.h ieeeep.h      portif.h
HAN@openindiana:~$

```



```

00007FFFAF2E0000    1524K r-x-- /lib/amd64/libc.so.1
00007FFFAF46D000     48K rw--- /lib/amd64/libc.so.1
00007FFFAF479000     16K rw--- /lib/amd64/libc.so.1
00007FFFAF48F000      4K rws-  [ anon ]
00007FFFAF4A0000     64K rwx--  [ anon ]
00007FFFAF4C0000    128K rwx--  [ anon ]
00007FFFAF4EB000     80K r---- /usr/lib/locale/en_US.UTF-8/LC_COLLATE/LCL_D
ATA
00007FFFAF500000     24K rwx--  [ anon ]
00007FFFAF510000      4K rwx--  [ anon ]
00007FFFAF520000      4K rwx--  [ anon ]
00007FFFAF530000      4K rwx--  [ anon ]
00007FFFAF540000      4K rwx--  [ anon ]
00007FFFAF550000      4K rw---  [ anon ]
00007FFFAF560000      4K rw---  [ anon ]
00007FFFAF570000      4K rwx--  [ anon ]
00007FFFAF580000      4K rwx--  [ anon ]
00007FFFAF590000      8K r--s-  [ comm ]
00007FFFAF599000    324K r-x-- /lib/amd64/ld.so.1
00007FFFAF5FA000     12K rwx-- /lib/amd64/ld.so.1
00007FFFAF5FD000      8K rwx-- /lib/amd64/ld.so.1
00007FFBFFFA000     24K rw---  [ stack ]
total                4588K
HANA@openindiana:~$

```

12. Over all personal experience

This project was about installing and using OpenIndiana in a virtual machine using Oracle VM VirtualBox. It helped to understand how operating systems are installed, configured, and managed in a safe virtual environment.

During the work, I learned basic system setup, virtualization concepts, and how UNIX-based systems operate. I also faced and solved common installation issues, which improved my practical skills.

OpenIndiana uses the ZFS filesystem and follows UNIX standards, making it stable and reliable for learning and system practice.

13. References

- OpenIndiana Official Documentation – <https://www.openindiana.org>
- Oracle VM VirtualBox User Guide – <https://www.virtualbox.org/manual>

- [UNIX System and POSIX Standards Overview – IEEE Standards Association](#)